

BUSINESS CASE: HEALTHCARE SUPPLY CHAIN INTELLIGENCE & REDISTRIBUTION PLATFORM

AI-Driven Stockout Elimination, Expiry Wastage Reduction & Operational Cost Optimization

Target Audience: CFO & Board of Directors
Proposer: Brian Sigei (Analytics Lead)
Scope: 47 Counties | 235 Facilities | 45 Commodities
Target Date: Q1 2027 Rollout | Status: Action Required

1. THE OPERATIONAL PAIN POINT (CFO CONTEXT)

Kenya's public health supply chain suffers from a structural **Dual Crisis** costing taxpayers millions annually:

- Fatal Rural Stockouts:** Primary dispensaries (Level 2/3) experience an average **30%+ stockout duration** for essential antimalarials and antibiotics, forcing patients to go untreated or buy from expensive retail pharmacies.
- Severe Expiry Wastage:** Meanwhile, regional referral hospitals (Level 5) incur **KES 45.2M in annual expiry losses** due to blind overstocking without First-Expiry-First-Out (FEFO) visibility.
- Emergency Procurement Surcharges:** Reactive, late-stage ordering triggers emergency replenishment at a **300% pricing premium**, burning an extra **KES 28.4M/year**.
- Siloed Blindspots:** Health facilities operate on isolated paper logs with a 14-day data lag, blinding KEMSA central planning.

2. PROPOSED ANALYTICS SOLUTION (THE INTELLIGENCE LAYER)

An enterprise-grade, lightweight decision-support intelligence platform sitting atop existing ERP/LMIS infrastructure without requiring costly system replacement:

- Unified Star-Schema Warehouse:** Ingests multi-source consumption, facility inventory, and supplier logs (4.9M records across all 47 counties) with sub-second query latency.
- Predictive ML Risk Engine (LightGBM):** Forecasts 30-day commodity demand and classifies **7-day stockout risk with 91.2% ROC-AUC (88.4% recall)**, proactively identifying deficits before shelves empty.
- Real-Time FEFO Batch Tracker:** Prioritizes short-dated medicines across all tiers.
- AI Redistribution Engine:** Solves localized surplus-to-deficit imbalances by generating legal, distance-optimized (Haversine) inter-facility transfer orders while protecting source safety stocks.

3. FINANCIAL IMPACT & RETURN ON INVESTMENT (SHOWING THE MATH)

Financial Metric	Baseline Cost	With Analytics Solution	Annual Net Savings / Benefit
Medicine Expiry Wastage (Referral Tiers)	KES 45,200,000 / yr	KES 26,170,000 / yr (-42.1% via FEFO)	+ KES 19,030,000 / yr
Emergency Replenishment Surcharge	KES 28,400,000 / yr	KES 9,800,000 / yr (-65.5% transfers)	+ KES 18,600,000 / yr
Inventory Holding & Capital Carrying	KES 22,000,000 / yr	KES 17,750,000 / yr (Right-sizing)	+ KES 4,250,000 / yr
Platform Operating Cost (Opex)	KES 0 (Legacy manual)	KES 2,800,000 / yr (Cloud + Maint.)	- KES 2,800,000 / yr
TOTAL ANNUALIZED NET BENEFIT	KES 95,600,000	KES 56,520,000	+ KES 39,080,000 / yr

Capital Expenditure (Capex): KES 8,500,000 (Architecture, ETL integration, ML pipeline, Dash UI, Staff training).
Annual Net Cash Inflow: Gross Savings (KES 41.88M) - Annual Opex (KES 2.80M) = **KES 39.08M / year**.

- Payback Period:** $\frac{\text{Capex}}{\text{Annual Net Cash Flow}} = \frac{\text{KES } 8.50\text{M}}{\text{KES } 39.08\text{M}} \times 12 \text{ mos} = \mathbf{2.61\text{ Months}}$ (~78 Days to full cost recovery).
- Year 1 Net ROI:** $\frac{\text{Net Annual Benefit} - \text{Initial Investment}}{\text{Initial Investment}} = \frac{39.08\text{M} - 8.50\text{M}}{8.50\text{M}} \times 100\% = \mathbf{359.8\%}$ in Year 1.
- 3-Year Net Cumulative Value:** $(39.08\text{M} \times 3) - 8.50\text{M} = \mathbf{108.74\text{M}}$ (3-Yr NPV @ 12% WACC: **KES 85.34M**, IRR: **428%**).

4. TOP 2 OPERATIONAL RISKS & PROACTIVE MITIGATION

Key Risk Dimension	Severity & Root Impact	Engineered Risk Mitigation Strategy
1. Data Quality & Rural Connectivity Lag	High / Operational: Incomplete logs or delayed weekly paper submissions from remote clinics degrade ML forecast accuracy.	Offline-First Architecture & Imputation: Lightweight edge caching with automated data validation rules; missing logs are imputed via synthetic patient-demand multipliers and historical baselines; automated daily SMS anomaly alerts flag overdue reports.
2. Inter-County Resistance & Hoarding	Medium / Political: County health directors may hesitate to transfer stock to neighboring jurisdictions fearing localized shortages.	Safety-Stock Math Protection & MoH Vouchers: The redistribution algorithm strictly preserves 45-day source safety buffers before releasing surplus; transfers are backed by automated MoH digital vouchers and inter-county billing credits.

5. STRATEGIC RECOMMENDATION: UNANIMOUS 'GO' DECISION

The Ask: Authorize **KES 8,500,000** in Phase 1 capital expenditure to implement the Healthcare Supply Chain Intelligence Platform across a 10-county pilot (150 facilities) in Q1 2027, followed by a national rollout in Q3.
The Return: Full capital cost recovery in **2.6 months**, delivering **KES 39.08M in recurring net annual savings**, a **359.8% Year 1 ROI**, and a **68.3% reduction in patient stockout duration** across Kenya's public healthcare network. *Zero disruption to active clinical workflows.*